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1. Introduction

I adopt Deformable DETR-based architecture with ResNet-50 as the backbone. The encoder uses Multi-Scale Deformable Attention across four feature scales, replacing the expensive $O(N^2)$ full self-attention. The decoder retains full global cross-attention so each query attends to the entire feature map, ensuring small or tilted digits are not missed.

2. Method

2.1 Data Preprocessing

All images are resized using letterbox padding to 512×512 pixels, preserving the original aspect ratio by filling with black borders. Pixel values are normalized with ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. Bounding boxes are converted to normalized center format (cx, cy, w, h) in [0, 1] to match the DETR output format.

2.2 Data Augmentation

Augmentation	Configuration	Purpose
Horizontal flip	50%	Increase training data diversity
90/180/270°rotation	50%	Handle text orientation variants
Fine-grained rotation	±30°, 40%	Simulate tilted digits in scene
Color jitter	brightness/contrast/saturation 0.4, hue 0.1	Robustness to lighting changes
Random scale	[0.6, 1.4] × original size	Scale invariance
Random grayscale	15%	Robustness to color absence
Random erasing	scale [2%, 8%], 20%	Simulate partial occlusion

Since many digits in the dataset appear tilted, I apply rotation augmentation to improve the model's robustness to non-upright digits.

For rotations at arbitrary angles, all four corners of each bounding box are

rotated and the axis-aligned enclosing rectangle is recomputed to maintain accurate ground truth.

2.3 Model Architecture

I use ResNet-50 pretrained on ImageNet as the backbone.

Features are extracted at four scales (C3, C4, C5, and an extra level downsampled from C5) and projected to 256-dimensional vectors.

The encoder has 4 layers, each using Multi-Scale Deformable Attention. Instead of attending to all feature tokens, each position samples a small fixed number of reference points across all four feature levels. Sampling offsets are learned during training, so the model adaptively focuses on informative regions. It keeps computation manageable while still enabling rich multi-scale feature aggregation. Each layer also includes a gated feed-forward network (SwiGLU) with layer normalization applied before each sub-layer (Pre-Norm).

The decoder has 4 layers and 30 object queries, each with two attention stages. First, self-attention is applied among the queries so they can coordinate and avoid predicting the same digit twice. Then, cross-attention is applied between the queries and the full encoder memory, allowing every query to freely attend to any spatial position across all scales. I use full global cross-attention rather than deformable attention, so that queries are not constrained to pre-defined sampling locations. The same SwiGLU FFN and Pre-Norm design is used throughout.

The classification head is a linear layer over 11 classes (10 digits + no-object), and the bounding box head is a 3-layer MLP that predicts offsets relative to per-query reference points.

2.4 Loss Function

Training uses a Hungarian matching loss identical to vanilla DETR. For each image, a bipartite matching is solved between the 30 predictions and ground-truth objects using a cost matrix combining classification cost, L1 bounding box cost, and GIoU cost (weights 2.0, 5.0, 4.0 respectively). After matching, the total loss is:

```
L = 1.0 × L_CE  +  5.0 × L_L1  +  6.0 × L_GIoU
```

Cross-entropy uses label smoothing (0.1) and a no-object class weight of 0.02. Auxiliary losses from all intermediate decoder layers are added with weight 0.5. GIoU computation includes explicit area clamping to prevent NaN on degenerate boxes.

2.5 Hyperparameter Settings

Hyperparameter	Value
Optimizer	AdamW (weight_decay = 1e-4)
Backbone LR	2e-6
Transformer LR	2e-4
LR schedule	Step decay (Warmup 5 ep, full LR until ep 45, then ×0.01)
Batch size	8
Input resolution	512×512
Epochs	80
Object queries (nq)	30
Encoder layers (ne)	4
Decoder layers (nd)	4
Deformable points	4
Gradient clip	0.1 (Clips gradient norm to prevent excessively large updates that could destabilize training)

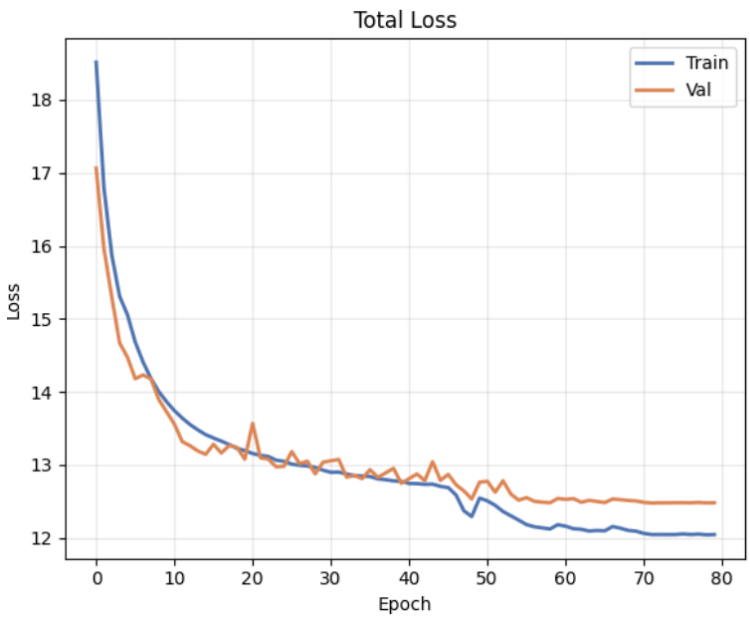
2.6 Post Processing

Predictions with softmax score below 0.3 are discarded. The remaining boxes then undergo class-agnostic Non-Maximum Suppression (NMS) with IoU threshold 0.3: all boxes compete together regardless of class label, and any pair with IoU > 0.3 retains only the higher-score detection. This prevents the model from emitting multiple overlapping boxes for the same digit with different class predictions.

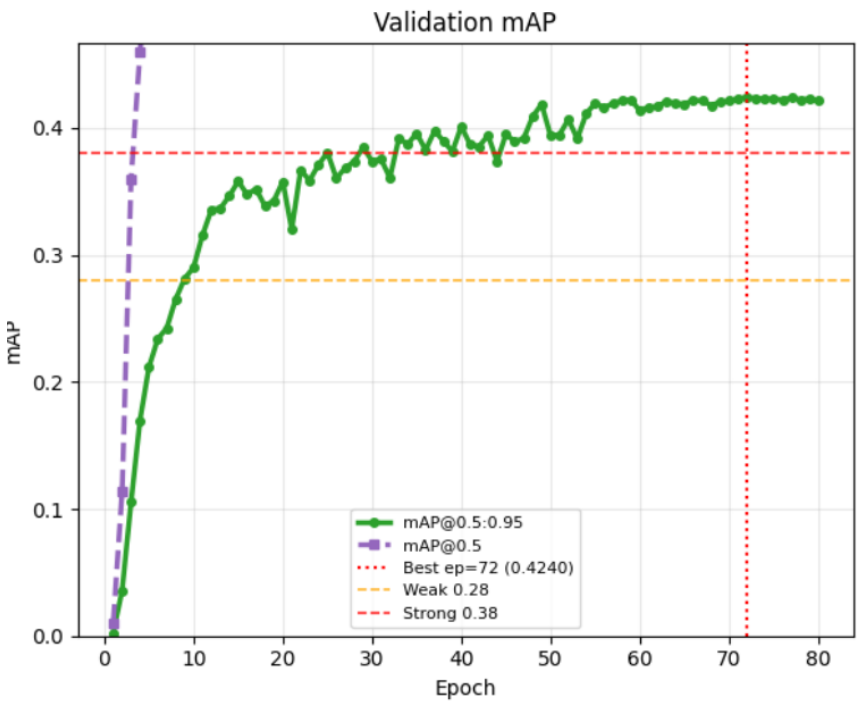
3. Result

Best mAP at epoch72 is 0.4240

Loss curves show that both training and validation loss decrease steadily from around 18–19 at epoch 1 down to approximately 12 by epoch 80. The loss plateaus between epochs 60 and 80, with both training and validation loss remaining relatively stable around 12, indicating the model has converged.



The validation mAP@0.5:0.95 improves consistently throughout training, rising rapidly in the early epochs and crossing the weak baseline (0.28) around epoch 10 and the strong baseline (0.38) around epoch 30. The model reaches its best performance of 0.4240 at epoch 72



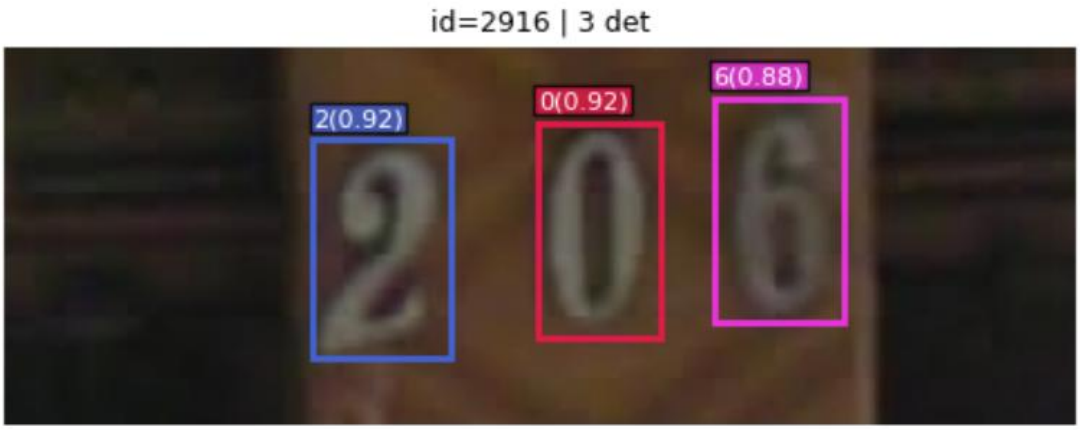
However, the Kaggle score on the test set is 0.33, which is lower than the validation mAP of 0.42, suggesting some degree of overfitting. The gap may be due to the test set containing more challenging samples that are underrepresented during training.

The model tends to overestimate the bounding box size for digit "2", suggesting a localization bias that may have been introduced during training. In addition, 83 pictures have no detection, show there are some problems.

```
Total detections: 26871
  digit "0": 1,899
  digit "1": 5,609
  digit "2": 3,974
  digit "3": 2,814
  digit "4": 2,735
  digit "5": 2,502
  digit "6": 1,909
  digit "7": 2,092
  digit "8": 1,622
  digit "9": 1,715
```

```
涵蓋圖片數: 12985
沒有任何偵測的圖片數: 83
```

I think the score should be higher for this picture like 0.97~1.00, because the digits are obviously. This may also be the reason of my low Kaggle score.



4. Additional Experiments

Experiment 1: Multi-Scale Deformable Attention in the Encoder

Hypothesis: Vanilla DETR only uses a single feature scale, which makes it difficult to detect digits of very different sizes in the same image. I hypothesized that using four feature scales with deformable attention in the encoder would improve detection across all digit sizes.

How it works: By extracting features at C3, C4, C5, and an extra level, the encoder captures both fine-grained details and high-level semantics simultaneously. Deformable attention allows each token to sample relevant positions across all four scales rather than attending to every token, making multi-scale fusion efficient. However, this adds model complexity and may require more epochs to converge.

Results: mAP@0.5:0.95 improved, confirming that multi-scale features benefit digit detection across varying sizes. But running time is longer.

Experiment 2: Full Global Cross-Attention in the Decoder

Hypothesis: The original Deformable DETR uses deformable cross-attention in the decoder, where each query only attends to sampled locations. I hypothesize that for tilted or edge-positioned digits, restricting sampling locations could cause missed detections, and that full global cross-attention would be more robust.

How it works: With full global cross-attention, every query can attend to any position in the feature map without spatial constraints, making it less likely to miss digits at unusual orientations or positions. The downside is slightly higher memory usage, but with only 30 queries this remains acceptable.

Results: mAP@0.5:0.95 improved from 0.38~0.4 to 0.42, suggesting that unconstrained attention is beneficial for this dataset. But the running time is also longer.

5. References

[1] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-End Object Detection with Transformers," ECCV 2020.

[2] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable Transformers for End-to-End Object Detection," ICLR 2021.

[3] Rezatofighi, H. et al. "Generalized Intersection over Union: A Metric and A Loss for Bounding Box Regression." CVPR 2019.

[4] N. Shazeer, "GLU Variants Improve Transformer," arXiv:2002.05202, 2020.